Newton School of Technology

# Data Visualization & Analytics

## Introduction to Capstone 2

***Policy + Rules + Structure***

# Capstone Introduction

Capstone 2 is an advanced, industry-style data analytics simulation. Student teams solve real-world sectoral problems using Python for the full ETL pipeline and Tableau for visualization and storytelling. The objective is to demonstrate end-to-end analytical capability with a focus on code quality, reproducibility, and decision-oriented insight delivery.

Each team selects a sector and a problem statement, then executes a complete analytics workflow — dataset sourcing, Python-based ETL, exploratory and statistical analysis, Tableau dashboard design, and final business recommendations. All work is version-controlled via a public GitHub repository using a structured branching and pull request workflow.

## What Changes from Capstone 1

| Area | Capstone 1 | Capstone 2 |
|---|---|---|
| ETL & Cleaning | Google Sheets | Python (Pandas, NumPy) |
| Statistical Analysis | Pivot Tables, basic formulas | Python (SciPy, Statsmodels) |
| Dashboarding | Google Sheets + Looker Studio | Tableau Public |
| Version Control | Google Sheet Version History | GitHub (branches, PRs, commits) |
| Contribution Audit | Sheet edit history | GitHub Insights + PR history |
| Submission Assets | PDF + PPT + Sheet link | 1. Report PDF<br>2. Project PPT<br>3. GitHub repo<br>4. Tableau Public URL<br>5. DVA-oriented-Resume<br>6. DVA-focused-Portfolio<br>(for reference: Profile Demo) |

## Stack Overview

| Tool | Status | Description |
|---|---|---|
| **Python** | Mandatory | All ETL, cleaning, and statistical analysis must be performed in Python using Jupyter Notebooks. Key libraries: pandas, numpy, matplotlib, seaborn, scipy, statsmodels. |
| **Google Colab** | Supported | Teams may use Google Colab as their Jupyter Notebook environment. All notebooks must still be committed to the GitHub repository in .ipynb format. |
| **Tableau Public** | Mandatory | All dashboards must be built and published on Tableau Public. Teams share the accessible public URL and submit screenshots in the GitHub repository. |
| **GitHub** | Mandatory | Each team maintains a public GitHub repository. All members contribute via branches and pull requests. Contribution is audited through GitHub Insights and PR history. |
| **SQL** | Optional | Teams may use SQL for initial data extraction if needed. Any SQL used must be documented in the repo. |

# Team Structure

- Team Size: 6 to 7 members per team
- Formation: Assigned by faculty. No swapping permitted.
- Requirement: Every member must have visible GitHub commits and PR contributions.

| Role | Primary Responsibility |
|---|---|
| Project Lead | Repo setup, timeline management, submission compliance, Gate 1 |
| Data Lead | Dataset sourcing, raw data commit, data dictionary in docs/ |
| ETL Lead | Notebooks 01 and 02 — extraction and cleaning pipeline |
| Analysis Lead | Notebooks 03 and 04 — EDA and statistical analysis |
| Visualization Lead | Tableau dashboard design, publishing, URL and screenshots |
| Strategy Lead | Problem statement, KPI framework, business recommendations |
| PPT & Quality Lead | Final report PDF, presentation deck, contribution matrix |

## GitHub Repository Structure

All teams must maintain the following folder and file structure in their public GitHub repository:

```
SectionName_TeamID_ProjectName/
- README.md
- data/
    - raw/        ← Original dataset (never edited)
    - processed/  ← Cleaned output
- notebooks/
    - 01_extraction.ipynb
    - 02_cleaning.ipynb
    - 03_eda.ipynb
    - 04_statistical_analysis.ipynb
    - 05_final_load_prep.ipynb
- scripts/
    - etl_pipeline.py
- tableau/
    - screenshots/
    - dashboard_links.md
- reports/
    - project_report.pdf
    - presentation.pdf
- docs/
    - data_dictionary.md

- DVA-oriented-Resume
- DVA-focused-Portfolio
```

# Capstone Workflow

| Step | Phase | Description |
|---|---|---|
| 1 | Define | Select sector, define problem statement, get mentor approval (Gate 1) |
| 2 | Extract | Source dataset, commit raw data to GitHub, document data dictionary |
| 3 | Clean & Transform | Python ETL pipeline — log all steps in notebooks |
| 4 | Analyze | EDA, statistical analysis, KPI computation in Python |
| 5 | Visualize | Build and publish Tableau dashboard, share URL and screenshots |
| 6 | Recommend | Deliver 3–5 actionable business recommendations tied to insights |
| 7 | Report | Final project report, presentation deck, contribution matrix |

# Evaluation Rubric (100 Marks)

| Area | Marks | Focus |
|---|---|---|
| Problem Framing | 10 | Is the business question clear and well-scoped? |
| Data Quality & ETL | 15 | Is the Python pipeline thorough and documented? |
| Analysis Depth | 25 | Are statistical methods applied correctly with insight? |
| Dashboard & Visualization | 20 | Is the Tableau dashboard interactive and decision-relevant? |
| Business Recommendations | 20 | Are insights actionable and well-reasoned? |
| Storytelling & Clarity | 10 | Is the presentation professional and coherent? |

*Remember: Marks are awarded for analytical thinking and decision relevance — not chart quantity, visual decoration, or code length.*